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הפקולטה למדעי ההנדסה

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כימיה של המים בהנדסה כימית סביבתית

Modeling the Aquatic Carbonate System and Water Acidification

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Submitted by

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1. Background

The carbonate system ($\text{CO}_2/\text{H}_2\text{CO}_3^*$, HCO_3^- , CO_3^{2-}) governs the pH and buffering capacity of nearly all natural waters. Dissolving CO_2 forms carbonic acid and lowers pH, the mechanism of ocean and freshwater acidification, while the same equilibria decide whether calcium carbonate precipitates (scaling) or dissolves (corrosion). We model how atmospheric CO_2 , alkalinity, temperature and ionic strength jointly control pH, total inorganic carbon and calcite saturation.

2. Methods

We built a code-based equilibrium model (provided as an interactive HTML tool and an equivalent Python/Matplotlib script). Inputs: CO_2 partial pressure $p\text{CO}_2$ (open system) or total inorganic carbon C_t (closed system), total alkalinity, temperature, ionic strength I , and total calcium. Alkalinity and hardness are reported both in meq/L and in the common “mg/L as CaCO_3 ” convention (multiply eq/L by 50,000 and mol/L by 100,000, respectively; e.g. 2 meq/L = 100 mg/L as CaCO_3).

Equations. The model uses the two carbonic-acid dissociation constants K_1 , K_2 , Henry’s constant K_h for CO_2 , the water ion product K_w , and the calcite solubility product K_{sp} . All five are evaluated as functions of temperature with the Plummer & Busenberg (1982) polynomials. Non-ideality is treated with the Davies equation, whose Debye-Hückel A-parameter is computed from the temperature-dependent dielectric constant of water; this converts thermodynamic constants into conditional (concentration) constants K_1' , K_2' , K_w' . The proton condition is expressed through alkalinity:

$$\text{Alk} = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + [\text{OH}^-] - [\text{H}^+]$$

Algorithm. For a given alkalinity the model finds the hydrogen-ion concentration $[\text{H}^+]$ that satisfies this balance by geometric bisection over pH 1-13 (200 iterations). In the open system $[\text{H}_2\text{CO}_3^*]$ is fixed by Henry’s law ($[\text{H}_2\text{CO}_3^*] = K_h \cdot p\text{CO}_2$) and the other species follow from K_1' , K_2' ; in the closed system the fixed C_t is distributed over the three species via the ionization fractions α_0 , α_1 , α_2 .

Outputs. Equilibrium pH (activity scale), the concentrations of H_2CO_3^* , HCO_3^- and CO_3^{2-} , total inorganic carbon C_t , and the calcite saturation state, reported as the saturation index $\text{SI} = \log_{10}(a_{\text{Ca}} \cdot a_{\text{CO}_3} / K_{sp})$ and the saturation ratio Ω . $\text{SI} > 0$ indicates over-saturation (precipitation potential); $\text{SI} < 0$ indicates under-saturation (dissolution / corrosivity). We also output the CCPP (Calcium Carbonate Precipitation Potential): the mass of CaCO_3 , in mg/L as CaCO_3 , that must precipitate (+) or dissolve (–) to reach equilibrium ($\Omega = 1$), a standard index for water stability and corrosion control (for example, desalinated RO water has a strongly negative CCPP and must be remineralized before distribution, whereas hard water has a positive CCPP and tends to scale). The model was validated against literature constants at 25 °C ($\text{p}K_1 = 6.35$, $\text{p}K_2 = 10.33$, $\text{p}K_h = 1.47$, $\text{p}K_{sp} = 8.48$, $A = 0.51$).

Dynamic extension (kinetics + mass transfer + multiple phases). Beyond the instantaneous equilibrium, the tool integrates a time-dependent scenario that couples three phases and two rate processes: (i) gas/liquid CO_2 mass transfer, $J = k_l a \cdot (K_h \cdot p\text{CO}_{2,\text{atm}} - [\text{CO}_2(\text{aq})])$, which changes C_t but not alkalinity, and (ii) calcite precipitation kinetics, $R = k_p \cdot (\Omega - 1)$, which removes Ca and consumes alkalinity ($\Delta\text{alk} = -2R$, $\Delta C_t = -R$). The coupled ordinary differential equations are integrated explicitly in time.

3. Results and Discussion

All curves below use a carbonate-buffered freshwater as the base case ($\text{Alk} = 2.0 \text{ meq/L}$, $\text{Ca} = 0.8 \text{ mM}$, $T = 15^\circ\text{C}$, $I = 0.01 \text{ M}$) unless stated. Figures 1-4 are the equilibrium results (one per group member); Figure 5 is an advanced dynamic result showcasing model complexity.

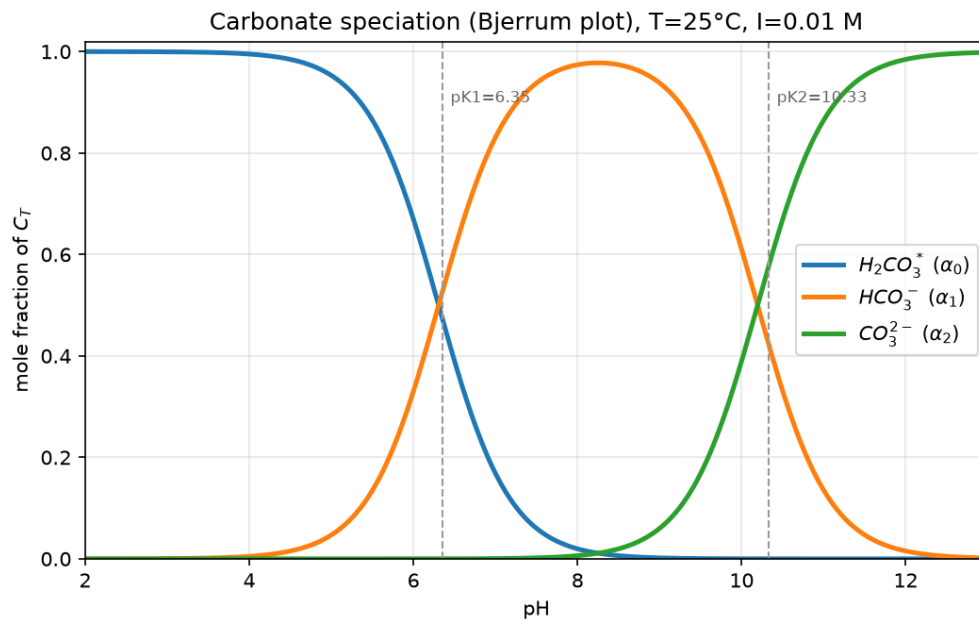


Figure 1. Bjerrum plot: carbonate speciation as a function of pH.

The relative abundance of the three carbonate species is fixed by pH. H_2CO_3^* dominates below pK_1 (6.35), HCO_3^- dominates between the two pK values, and CO_3^{2-} takes over above pK_2 (10.33); the curves cross exactly at the pK values, where the two neighbouring species are equal. Around the circum-neutral pH of most natural waters (7-8.5) bicarbonate carries essentially all of C_T , which is why HCO_3^- is the principal contributor to alkalinity and to the buffering capacity of the system.

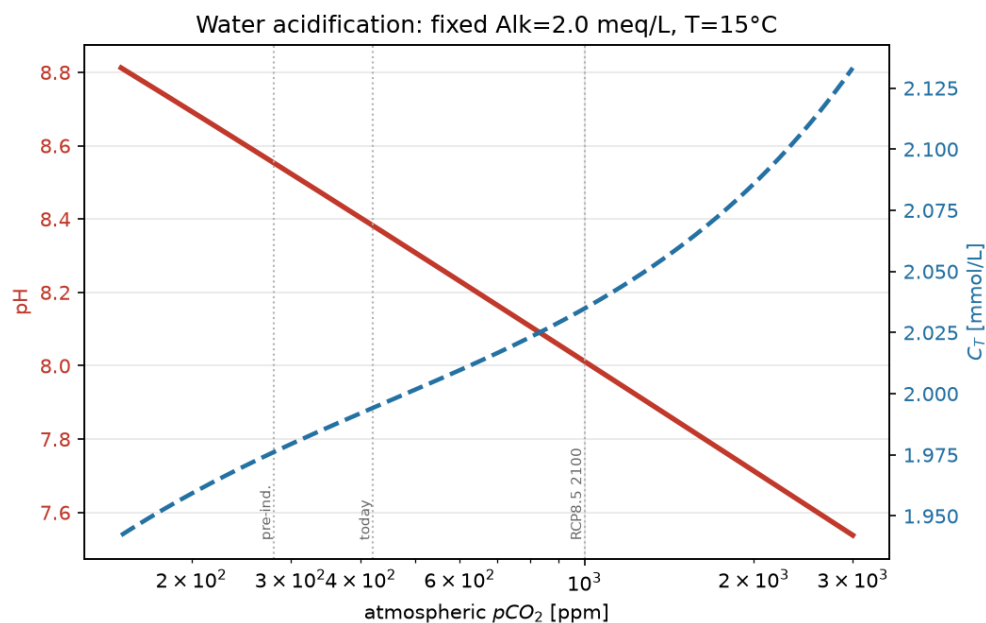


Figure 2. Acidification: equilibrium pH (red) and C_T (blue) versus atmospheric pCO_2 at constant alkalinity.

Raising pCO_2 drives more CO_2 into solution (Henry's law); the added H_2CO_3^* dissociates and releases H^+ , so pH falls almost linearly with $\log \text{pCO}_2$, the essence of ocean and

freshwater acidification. Because alkalinity is conserved, the extra acidity is buffered by converting CO_3^{2-} and HCO_3^- , and total inorganic carbon C_t rises. Moving from the pre-industrial (280 ppm) to a high-emission 2100 level (~ 1000 ppm) lowers the modelled pH by several tenths of a unit, matching observed trends.

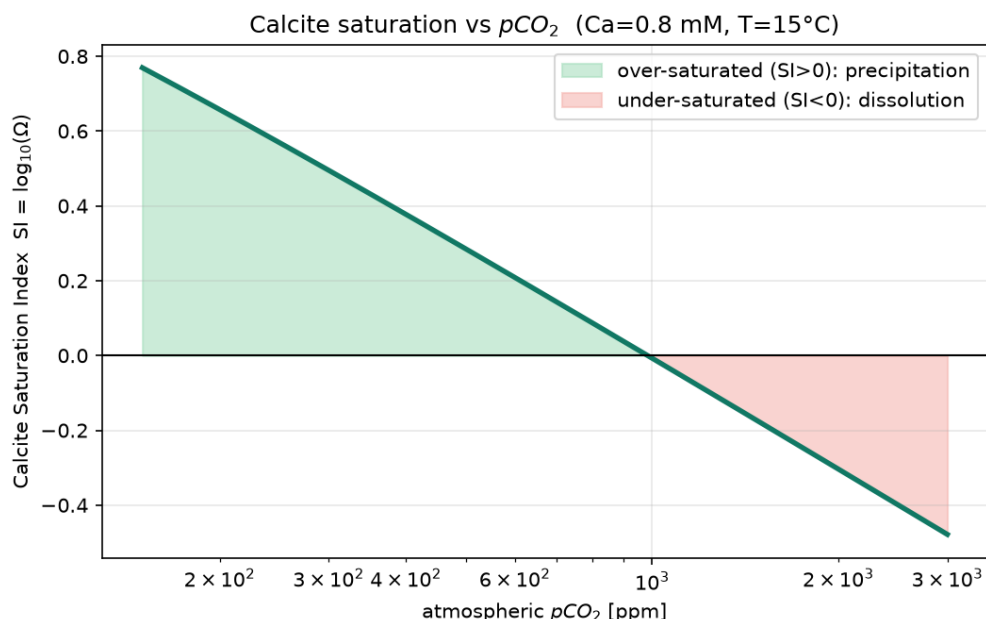


Figure 3. Calcite saturation index versus $p\text{CO}_2$, showing the over/under-saturation threshold.

The saturation index falls as $p\text{CO}_2$ rises, because acidification lowers pH and shifts speciation away from CO_3^{2-} , reducing the ion-activity product $a_{\text{Ca}} \cdot a_{\text{CO}_3}$. The curve crosses $\text{SI} = 0$ near ≈ 1000 ppm: to the left the water is over-saturated (green) and calcite tends to precipitate; to the right it becomes under-saturated (red) and calcite dissolves, i.e. the water turns corrosive to carbonate minerals and cement. This single threshold links climate-scale CO_2 to practical concerns of scaling versus corrosion in pipes and aquifers.

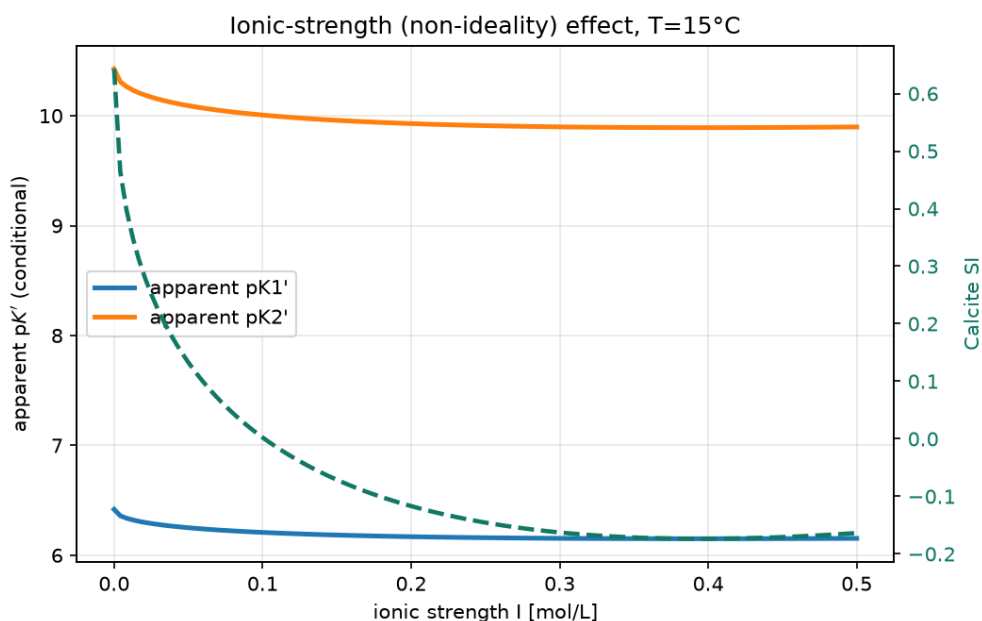


Figure 4. Effect of ionic strength on the apparent pK values and calcite SI .

Increasing ionic strength lowers activity coefficients (Davies equation), which lowers the apparent pK_1' and pK_2' , dissolved ions are effectively “shielded”, so the acids appear stronger. The divalent CO_3^{2-} is affected more than the monovalent species, so pK_2' shifts most. Calcite SI first drops sharply then levels off, because the doubly-charged Ca^{2+} and CO_3^{2-} activities are strongly reduced at higher I. This explains why saline waters can hold much more dissolved carbonate before precipitating, and why ignoring non-ideality overestimates scaling potential.

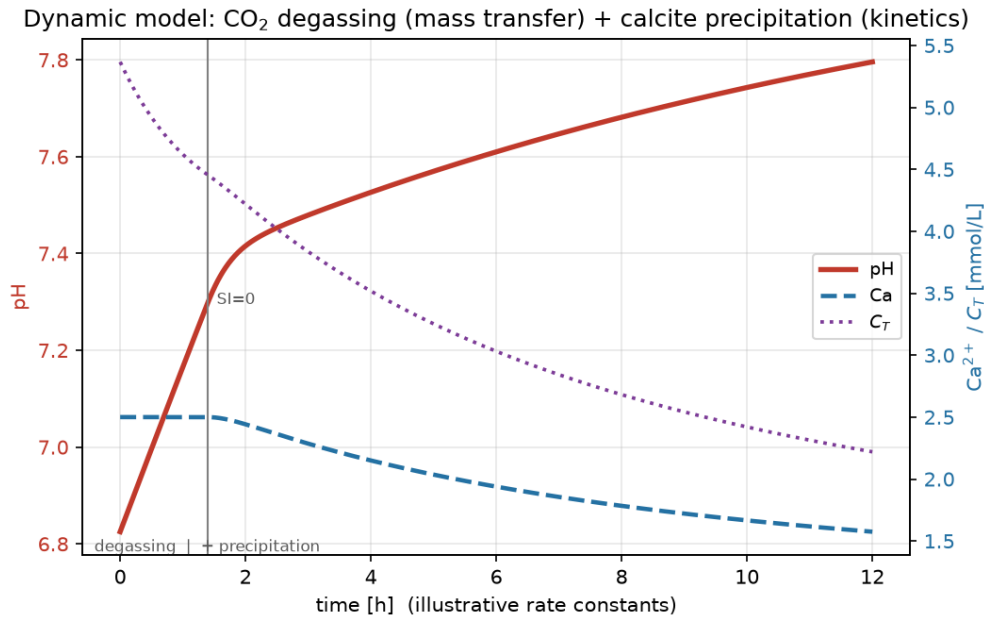


Figure 5 (advanced). Dynamic model coupling CO_2 mass transfer and calcite precipitation kinetics across three phases.

A CO_2 -rich groundwater (equilibrated with high soil $p\text{CO}_2$) is exposed to the atmosphere. First, CO_2 degasses by gas/liquid mass transfer: C_t falls and pH rises steeply while Ca stays constant. When the water crosses into super-saturation ($\text{SI} = 0$, marked), calcite precipitation kinetics switch on and Ca drops as CaCO_3 forms, consuming alkalinity. The system self-regulates near $\text{SI} \approx 0$ as precipitation tracks continued degassing. This reproduces travertine and pipe-scale formation, and simultaneously demonstrates kinetics, mass transfer and multiple phases.

4. Conclusions

A single set of carbonate equilibria, solved through the alkalinity balance, reproduces water acidification, buffering, and the precipitation-dissolution behaviour of calcite. The tool quantifies how CO_2 , alkalinity, temperature and ionic strength move a water across the $\text{SI} = 0$ line, directly useful for predicting acidification impacts, corrosion control, carbonate scaling in desalination and cooling systems, and chemical stabilization of treated water.